

Equations for NE533

$$Q = E_f \times N_f \times \sigma_f \times \phi$$

$$T(r) = \frac{Q(R^2 - r^2)}{4k} + T_s$$

$$T_0 - T_s = \frac{Q}{4k} R_f^2 = \frac{LHR}{4\pi k}$$

$$T_s - T_{IC} = \frac{Q}{2h_{gap}} R_f = \frac{LHR}{2\pi R_f h_{gap}}$$

$$T_{IC} - T_{OC} = \frac{Q}{2h_{clad}} R_f = \frac{LHR}{2\pi R_f h_{clad}}$$

$$T_{OC} - T_{cool} = \frac{Q}{2h_{cool}} R_f = \frac{LHR}{2\pi R_f h_{cool}}$$

$$h_{gap/clad} = \frac{k_{gap/clad}}{t_{gap/clad}}$$

$$LHR \left(\frac{z}{Z_0} \right) = LHR^0 \cos \left[\frac{\pi}{2\gamma} \left(\frac{z}{Z_0} - 1 \right) \right]$$

$$T_{cool}(z) - T_{cool}^{in} = \frac{2\gamma}{\pi} \frac{Z_0 \times LHR^0}{\dot{m} C_p} \times$$

$$\left\{ \sin \frac{\pi}{2\gamma} + \sin \left[\frac{\pi}{2\gamma} \left(\frac{z}{Z_0} - 1 \right) \right] \right\}$$

$$\sigma_\theta = \frac{pR}{\delta} \quad \sigma_z = \frac{pR}{2\delta} \quad \sigma_r = -\frac{p}{2}$$

$$\sigma_r(r) = -p \frac{(R_o/r)^2 - 1}{(R_o/R_i)^2 - 1}$$

$$\sigma_\theta(r) = p \frac{(R_o/r)^2 + 1}{(R_o/R_i)^2 - 1}$$

$$\sigma_z(r) = p \frac{1}{(R_o/R_i)^2 - 1}$$

$$\sigma = -E\Delta T\alpha$$

$$\sigma_r(r) = \frac{1}{2}\Delta T \frac{\alpha E}{1-\nu} \left(\frac{r}{R_i} - 1 \right) \left(1 - \frac{R_i}{\delta} \left(\frac{r}{R_i} - 1 \right) \right)$$

$$\sigma_\theta(r) = \frac{1}{2}\Delta T \frac{\alpha E}{1-\nu} \left(1 - 2\frac{R_i}{\delta} \left(\frac{r}{R_i} - 1 \right) \right)$$

$$\sigma_z(r) = \frac{1}{2}\Delta T \frac{\alpha E}{1-\nu} \left(1 - 2\frac{R_i}{\delta} \left(\frac{r}{R_i} - 1 \right) \right)$$

$$\sigma_r(\eta) = -\sigma^*(1 - \eta^2)$$

$$\sigma_\theta(\eta) = -\sigma^*(1 - 3\eta^2)$$

$$\sigma_z(\eta) = -2\sigma^*(1 - 2\eta^2)$$

$$\sigma^* = \frac{\alpha E(T_0 - T_s)}{4(1 - \nu)}$$

$$\Delta\delta_{gap} = \bar{R}_c\alpha_c(\bar{T}_c - T_{ref}) - R_f\alpha_f(\bar{T}_f - T_{ref})$$

$$D(t)^2 - D_0^2 = 2M_{GB}\gamma_{GB}t$$

$$M_{GB}(T) = M_0 e^{(-\frac{Q}{k_B T})}$$

$$\frac{dD}{dt} = k \left(\frac{1}{D} - \frac{1}{D_m} \right)$$

$$\epsilon_D = \Delta\rho_0 \left(e^{\frac{\beta I n_{0.01}}{C_D \beta_D}} - 1 \right)$$

$$\beta = \frac{\dot{F}t}{N_U}$$

Post-irradiation Annealing

$$\tau < \pi^{-2} \rightarrow f = 6\sqrt{\frac{Dt}{\pi a^2}} - 3\frac{Dt}{a^2}$$

$$\tau \geq \pi^{-2} \rightarrow f = 1 - \frac{6}{\pi^2} \exp(-\pi^2 \frac{Dt}{a^2})$$

In-pile gas release

$$\tau < \pi^{-2} \rightarrow f = 4\sqrt{\frac{Dt}{\pi a^2}} - \frac{3}{2} \frac{Dt}{a^2}$$

$$\tau \geq \pi^{-2} \rightarrow f = 1 - \frac{0.662}{\frac{Dt}{a^2}} \left(1 - 0.92 \exp(-\pi^2 \frac{Dt}{a^2}) \right)$$

$$D = D_1 + D_2 + D_3 \text{ cm}^2/\text{s}$$

$$D_1 = 7.6 \times 10^{-6} \exp\left(\frac{-3.03}{k_B T}\right)$$

$$D_2 = 1.41 \times 10^{-18} \exp\left(\frac{-1.19}{k_B T}\right) \sqrt{\dot{F}}$$

$$D_3 = 2.0 \times 10^{-30} \dot{F}$$

$$\epsilon_{SFP} = 5.577 \times 10^{-2} \rho \beta$$

$$\epsilon_{GFP} = 1.96 \times 10^{-28} \rho \beta (2800 - T)^{11.73} \times$$

$$e^{-0.0162(2800-T)} e^{-17.8\rho\beta}$$

$$\dot{\epsilon}_{th} = A_0 \left(\frac{\sigma}{G} \right)^n e^{(\frac{-Q}{RT})}$$

$$\dot{\epsilon}_{ir} = C_0 \Phi^{C_1} \sigma^{C_2}$$

$$\Phi = 3E11 \times LHR$$